STPM 2025 PHYSICS 960/1 (U1)

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.

Section A [15 marks]

1. The measurements of length, diameter and mass of a solid cylinder are (12.0 ± 0.2) cm, (5.0 ± 0.1) cm and (1.54 ± 0.02) kg respectively. What is the density of the cylinder?

A $(6.5 \pm 0.5) \times 10^3$ kg m^{-3} B $(6.5 \pm 0.1) \times 10^3$ kg m^{-3}
C $(1.6 \pm 0.1) \times 10^3$ kg m^{-3} D $(1.6 \pm 0.2) \times 10^3$ kg m^{-3}

Answer: A

$$V = \pi r^2 L = \pi \left(\frac{d}{2}\right)^2 L = \frac{\pi d^2 L}{4} = \frac{\pi(0.050)^2(0.120)}{4} = 2.36 \times 10^{-4}\ \text{m}^3$$

$$\rho = \frac{m}{V} = \frac{4m}{\pi d^2 L} = \frac{4(1.54)}{\pi(0.050)^2(0.120)} = 6.54 \times 10^3\ \text{kg m}^{-3}$$

$$\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \frac{2\Delta d}{d} + \frac{\Delta L}{L} = \frac{0.02}{1.54} + \frac{2(0.1)}{5.0} + \frac{0.2}{12.0} = 0.0697$$

$\Delta\rho = 0.0697 \times 6.54 \times 10^3 \approx 0.5 \times 10^3$ kg m^{-3}, so $\rho = (6.5 \pm 0.5) \times 10^3$ kg m^{-3}.

Explanation: Density is $\rho = \frac{4m}{\pi d^2 L}$. When quantities are multiplied or divided, their *fractional* uncertainties add. Because d is squared, its fractional uncertainty is counted twice, and this term (0.04) is the largest contribution — that is why the uncertainty is $\pm 0.5 \times 10^3$, not $\pm 0.1 \times 10^3$. Options C and D come from using the diameter as if it were the radius.

2. A car travels at a uniform velocity of 17.3 m s^{-1}. The velocity of the car is then increased with a uniform acceleration of 21.2 m s^{-1} in 3.3 s. What is the distance travelled by the car while the car accelerates?

A 12.9 m B 63.5 m C 70.0 m D 127 m

Answer: B

The unit m s^{-1} shows the velocity increases from 17.3 m s^{-1} to 21.2 m s^{-1} in 3.3 s.

$$s = \frac{(u+v)t}{2} = \frac{(17.3 + 21.2)(3.3)}{2} = 63.5\ \text{m}$$

Explanation: Always check the unit: acceleration is measured in m s^{-2}, so "21.2 m s^{-1}" must be the velocity reached after 3.3 s. For uniform acceleration the average velocity is $\frac{u+v}{2}$, and distance = average velocity × time. (If 21.2 were treated as an acceleration, the distance would be 172 m, which is not among the options — a useful check.)

3. A vehicle, P, travels at constant velocity collides with another vehicle, Q, which stops suddenly. The masses of P and Q are the same. Which statement is true about Q if P stops after the collision?

A The momentum of Q after the collision is equal to the momentum of P before the collision.
B The momentum of Q after the collision is less than the momentum of P before the collision.
C The kinetic energy of Q after the collision is less than the kinetic energy of P before the collision.
D The kinetic energy of Q after the collision is equal to the kinetic energy of P before the collision.

Answer: A

By conservation of linear momentum, $m_P u_P + 0 = 0 + m_Q v_Q$, so all the momentum of P is transferred to Q.

Note: with equal masses, $v_Q = u_P$, so the kinetic energy is also unchanged (statement D also follows). A is the answer based on the principle of conservation of momentum, which always holds in a collision.

Explanation: During a collision the forces between P and Q are internal, so the total momentum of the system is conserved. Since P stops, all of its momentum is transferred to Q. Kinetic energy is conserved only in an elastic collision, so the statement that is always guaranteed is the one about momentum.

4. A block of mass 2.0 kg is placed on a rough surface of one horizontal plank which is then tilted gradually. The block starts to descend the plank when the inclination angle exceeds 20°. What is the coefficient of static friction between the plank and the block?

A 0.34 B 0.36 C 0.94 D 1.00

Answer: B

At the point of slipping, $mg \sin\theta = \mu_s mg \cos\theta$

$$\mu_s = \tan\theta = \tan 20° = 0.36$$

Explanation: Just before the block slips, static friction is at its maximum value $\mu_s R$ and balances the component of weight along the plank, $mg \sin\theta$. The normal reaction is $R = mg \cos\theta$, so m and g cancel and $\mu_s = \tan\theta$. The mass of 2.0 kg is not needed.

5. Each tyre of a car has a diameter of 0.360 m which makes 200 revolutions per minute while moving on a straight road. What is the speed of the car?

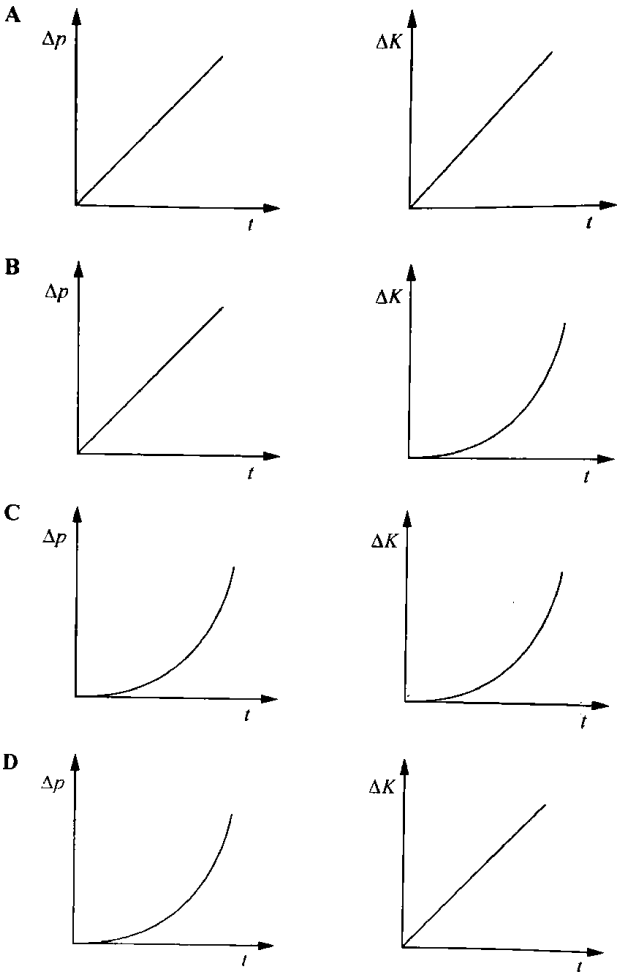
A 1.20 m s^{-1} B 1.88 m s^{-1} C 3.77 m s^{-1} D 7.54 m s^{-1}

Answer: C

$$v = \pi d \times f = \pi(0.360) \times \frac{200}{60} = 3.77 \text{ m s}^{-1}$$

Explanation: When a wheel rolls without slipping, the car moves forward one circumference (πd) for every revolution. Convert revolutions per minute into revolutions per second by dividing by 60. Option B uses the radius instead of the diameter; option D uses $2\pi \times$ diameter.

6. An object is accelerated from rest by a constant force for a period of time. Which graphs represent the variation of change in momentum, Δp, with time, t, and the variation of change in kinetic energy, ΔK, with time, t, of the object?



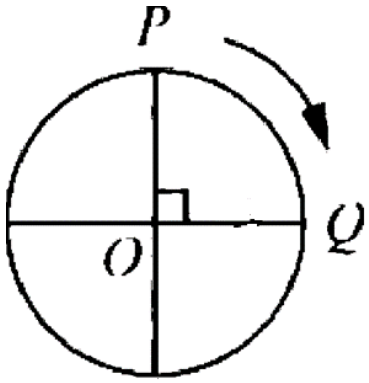
Answer: B

$\Delta p = Ft$, so $\Delta p \propto t$ (straight line through the origin).

$\Delta K = \frac{(\Delta p)^2}{2m} = \frac{F^2 t^2}{2m}$, so $\Delta K \propto t^2$ (parabola).

Explanation: Newton's second law: a constant force gives a constant rate of change of momentum, so $\Delta p = Ft$ is a straight line through the origin. Since $K = \frac{p^2}{2m}$, $\Delta K \propto t^2$, a curve that becomes steeper: the object moves faster each second, so the same force does more work per second.

7. A particle moves in a horizontal circle about the centre of the circle, O, with a radius of 0.50 m at a constant speed from P to Q in 2.0 s as shown in the diagram below.



Which statement is **not** true about the motion of the particle?

A The acceleration of the particle is 0.20 m s^{-2}.

B The linear speed of the particle is 0.39 m s^{-1}.

C The angular velocity of the particle from P to Q is 0.79 rad s^{-1}.

D The angular displacement of the particle from P to Q is $\frac{\pi}{2}$ rad.

Answer: A

$$\theta = \frac{\pi}{2} \text{ rad}$$

$$\omega = \frac{\theta}{t} = \frac{\pi}{2(2.0)} = 0.79 \text{ rad s}^{-1}$$

$$v = r\omega = 0.50(0.785) = 0.39 \text{ m s}^{-1}$$

$a = r\omega^2 = 0.50(0.785)^2 = 0.31 \text{ m s}^{-2} \neq 0.20 \text{ m s}^{-2}$, so A is not true.

Explanation: P to Q is one quarter of a circle, so the angular displacement is $\frac{\pi}{2}$ rad. Even at constant speed the direction of velocity keeps changing, so there is a centripetal acceleration $a = r\omega^2 = \frac{v^2}{r}$ towards O. Its value is 0.31 m s^{-2}, so statement A is the incorrect one.

8. A satellite of mass 20 kg moves from X to Y. The gravitational potential at X is −8.0 kJ kg^{-1} and the gravitational potential at Y is −4.0 kJ kg^{-1}. What is the change in the potential energy of the satellite when the satellite moves from X to Y?

A −160 kJ B −80 kJ C 80 kJ D 160 kJ

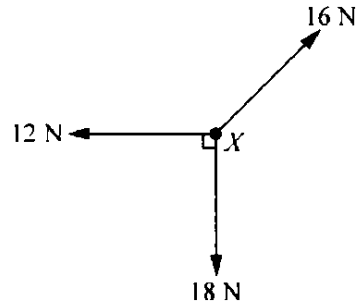
Answer: C

$$\Delta U = m(V_Y - V_X) = 20[-4.0 - (-8.0)] = +80 \text{ kJ}$$

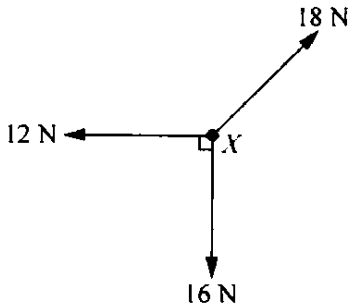
Explanation: Potential energy = mass × potential, $U = mV$. Moving from −8.0 kJ kg^{-1} to −4.0 kJ kg^{-1} means the potential becomes less negative (the satellite moves further from the Earth), so the potential energy *increases* — the change is positive. Option B has the wrong sign.

9. An object, *X*, is acted by three forces. Which diagram is correct when *X* is in equilibrium?

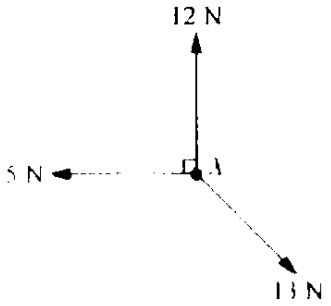
A



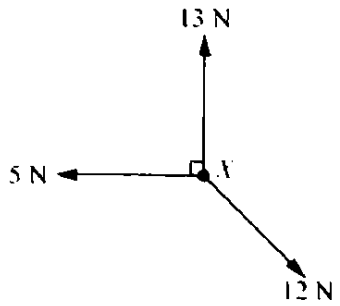
B



C



D



Answer: C

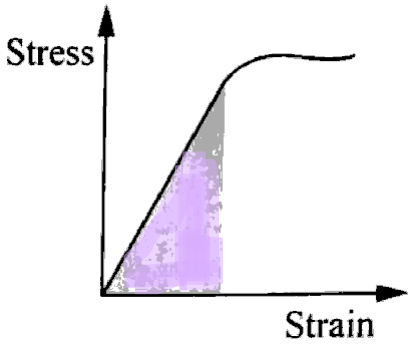
For equilibrium, the diagonal force must equal the resultant of the two perpendicular forces and act in the opposite direction.

C: $\sqrt{5^2 + 12^2} = 13$ N ✓ A: $\sqrt{12^2 + 18^2} = 21.6$ N $\neq$ 16 N B: $\sqrt{12^2 + 16^2} = 20$ N $\neq$ 18 N

D: $\sqrt{5^2 + 13^2} = 13.9$ N $\neq$ 12 N

Explanation: Three forces in equilibrium form a closed triangle: the resultant of any two forces must be equal in size and opposite in direction to the third force. The resultant of two perpendicular forces a and b is $\sqrt{a^2 + b^2}$. Only 5 N, 12 N and 13 N satisfy this ($5^2 + 12^2 = 13^2$), and the 13 N force points opposite to the resultant of the other two.

10. The variation of stress with strain for a metal wire is shown in the graph below.



Which is represented by the shaded area of the graph?

A The Young's modulus of the wire
B The strain energy stored in the stretched wire
C The work done to stretch the wire per unit volume
D The minimum work done to produce the permanent extension

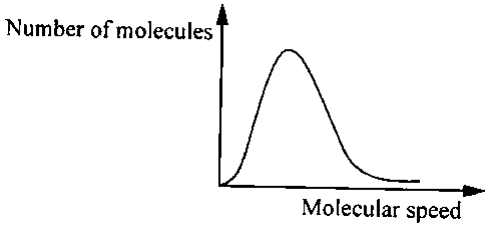
Answer: C

$$\text{Area} = \frac{1}{2}\sigma\varepsilon = \frac{1}{2}\left(\frac{F}{A}\right)\left(\frac{e}{l}\right) = \frac{\frac{1}{2}Fe}{Al} = \frac{W}{V}$$

i.e. work done (strain energy) per unit volume. The gradient, not the area, gives the Young's modulus.

Explanation: Stress × strain $= \frac{F}{A} \times \frac{e}{l} = \frac{Fe}{Al}$. Since Fe is related to work and Al is the volume of the wire, the area under a stress–strain graph is energy *per unit volume*. The total strain energy (B) is the area under a *force–extension* graph. The gradient of the linear part gives the Young's modulus (A).

11. The molecular speed distribution is shown in the graph below.



Which statement is **not** true about the graph?

A A large number of molecules moves at the highest speed.

B The largest number of molecules is at the most probable speed.

C The number of molecules with higher speed decreases rapidly than the number of molecules with the root mean square speed.

D The number of molecules with the average speed is more than the number of molecules with the root mean square speed.

Answer: A

The curve falls towards zero at high speeds, so only a very small number of molecules move at the highest speeds. (Since $c_{\text{mp}} < \bar{c} < c_{\text{rms}}$, statement D is true.)

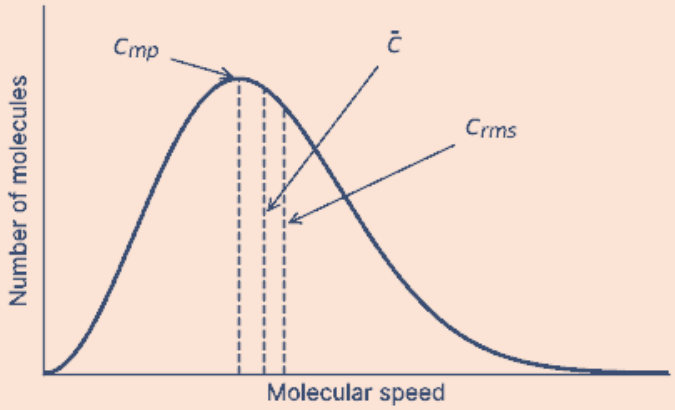
Explanation: What the graph shows. This is the Maxwell–Boltzmann distribution of molecular speeds for a gas at a fixed temperature. The vertical axis shows the number of molecules having a particular speed, and the total area under the curve equals the total number of molecules in the gas.

Key features of the curve

- The curve starts at zero: no molecule is completely at rest.
- It rises to a peak. The speed at the peak is the *most probable speed*, c_{mp} — more molecules have this speed than any other speed.
- After the peak the curve falls and has a long "tail" towards high speeds. It approaches the axis but only a very small number of molecules have very high speeds.
- Because of this long tail the curve is not symmetrical, so the three typical speeds are different:

$$c_{\text{mp}} = \sqrt{\frac{2kT}{m}} \quad < \quad \bar{c} = \sqrt{\frac{8kT}{\pi m}} \quad < \quad c_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

(in the ratio about 1 : 1.13 : 1.22)



Position of c_{mp}, $\bar{c}$ and c_{rms} on the distribution curve: c_{mp} is at the peak, $\bar{c}$ is slightly to the right, and c_{rms} is further right, on the falling side of the curve.

Checking each option

- **A — not true.** At the highest speeds the curve is very close to the axis, so only a *small* number of molecules move at the highest speed. This is the false statement, so the answer is **A**.
- **B — true.** The peak of the curve is at the most probable speed, so the largest number of molecules has this speed.
- **C — true.** To the right of the peak the curve slopes downward: for speeds greater than c_{rms} the number of molecules keeps decreasing rapidly.
- **D — true.** $\bar{c}$ is closer to the peak than c_{rms}, so the curve is higher at $\bar{c}$. Hence more molecules have the average speed than the r.m.s. speed.

Tip: When the temperature increases, the peak shifts to a higher speed and becomes lower and broader, but the total area under the curve (the number of molecules) stays the same.

12. Methane is a polyatomic gas. What is the average kinetic energy for a molecule of methane gas at 50.0 °C?

A 2.07×10^{-21} J B 6.69×10^{-21} J C 1.11×10^{-20} J D 1.34×10^{-20} J

Answer: D

A polyatomic gas molecule has 3 translational + 3 rotational = 6 degrees of freedom, each with energy $\frac{1}{2}kT$.

$$\overline{E} = \frac{6}{2}kT = 3(1.38\times 10^{-23})(323.15) = 1.34\times 10^{-20}\ \text{J}$$

Explanation: Principle of equipartition of energy. Each degree of freedom of a gas molecule has an average energy of $\frac{1}{2}kT$. So the average kinetic energy of one molecule is $E = \frac{f}{2}kT$, where f is the number of degrees of freedom.

Degrees of freedom of gas molecules (at normal temperatures)

- Monatomic gas (e.g. He, Ne): 3 translational, $f = 3$, $E = \frac{3}{2}kT$
- Diatomic gas (e.g. O_2, N_2): 3 translational + 2 rotational, $f = 5$, $E = \frac{5}{2}kT$
- Polyatomic gas (e.g. CH_4, H_2O): 3 translational + 3 rotational, $f = 6$, $E = \frac{6}{2}kT = 3kT$

A polyatomic molecule can rotate about all three axes, so it has one more rotational degree of freedom than a diatomic molecule.

Applying to this question. The question states that methane is a polyatomic gas, so $f = 6$ and $E = 3kT$ with $T = 50.0 + 273.15 = 323.15$ K, giving 1.34×10^{-20} J (D). Option B (6.69×10^{-21} J) is $\frac{3}{2}kT$, which is only the *translational* kinetic energy — a common mistake.

13. An ideal gas has a molar heat capacity at constant volume of C_V. If R is the molar gas constant, what is the ratio of the molar heat capacity at constant pressure to molar heat capacity at constant volume?

A $\frac{C_V - R}{C_V}$ B $\frac{C_V}{C_V - R}$ C $\frac{C_V}{C_V + R}$ D $\frac{C_V + R}{C_V}$

Answer: D

$$C_p - C_V = R \quad \Longrightarrow \quad \gamma = \frac{C_p}{C_V} = \frac{C_V + R}{C_V}$$

Explanation: At constant pressure the gas expands, so part of the heat supplied is used to do work. More heat is therefore needed for the same temperature rise, and $C_p = C_V + R$. The ratio $\gamma = \frac{C_p}{C_V}$ is always greater than 1.

14. The first law of thermodynamics can be written as $Q = \Delta U + W$, where Q is heat, ΔU is the change of internal energy and W is work. For an amount of a certain gas, heat is released by the gas and temperature decreases when the gas expands. Which is true about Q, ΔU and W?

	Q	ΔU	W
A	Negative	Negative	Negative
B	Negative	Negative	Positive
C	Positive	Positive	Negative
D	Positive	Positive	Positive

Answer: B

Heat released $\Longrightarrow$ $Q < 0$; temperature falls $\Longrightarrow$ $\Delta U < 0$; gas expands (work done by the gas) $\Longrightarrow$ $W > 0$.

Explanation: Sign convention for $Q = \Delta U + W$: Q is positive when heat is supplied to the gas, and W is positive when work is done *by* the gas. Heat released, so $Q < 0$. The internal energy of an ideal gas depends only on temperature, and the temperature falls, so $\Delta U < 0$. The gas expands, so it does work and $W > 0$.

15. Which statement is true about heat transfer?

A Thermal conductivity of a metal increases with temperature.
B Heat transfer only occurs between regions of the same temperature.
C Doubling the length of a conductor will double the rate of heat flow.
D Doubling the cross-sectional area of a conductor will double the rate of heat flow.

Answer: D

$\frac{dQ}{dt} = -kA\frac{d\theta}{dx}$, so the rate of heat flow is directly proportional to A (and inversely proportional to the length).

Explanation: The rate of heat flow is $\frac{dQ}{dt} = kA\frac{\Delta\theta}{L}$: it is directly proportional to the cross-sectional area A and inversely proportional to the length L. Doubling the length halves the rate (C is wrong). Heat flows only when there is a temperature difference (B is wrong).

Section B [15 marks]

16. A steel cable of length 2.50 m and diameter 30.0 mm is used to lift a 500 kg wooden crate. The Young's modulus of the steel is 2.00×10^{11} Pa.

(a) Determine the stress of the cable. [3]

Answer

$$A = \pi r^2 = \pi(0.0150)^2 = 7.07 \times 10^{-4} \text{ m}^2$$ 1

$$\sigma = \frac{F}{A} = \frac{mg}{A} = \frac{500(9.81)}{7.07 \times 10^{-4}}$$ 1

$= 6.94 \times 10^6$ Pa OR 6.9×10^6 Pa OR 6.939×10^6 Pa 1

3

Explanation: Stress is force per unit cross-sectional area. Here the force in the cable is the weight of the crate, *mg*. Use the radius (half the diameter) and convert mm to m before calculating the area.

(b) Calculate the extension of the cable. [3]

Answer

$E = \frac{\sigma}{\varepsilon} = \frac{\sigma l}{e}$ OR $e = \frac{\sigma l}{E}$ 1

$e = \frac{(6.94 \times 10^6)(2.50)}{2.00 \times 10^{11}}$ (ECF from (a)) 1

$= 8.67 \times 10^{-5}$ m OR 8.7×10^{-5} m OR 8.674×10^{-5} m 1

3

Explanation: Young's modulus links stress and strain: $E = \frac{\sigma}{\varepsilon}$ with strain $\varepsilon = \frac{e}{l}$. The extension is very small (about 0.09 mm) because steel is very stiff — this is exactly why steel is chosen for lifting cables.

(c) Steel is a suitable material for the cable. State *two* reasons for the suitability. [2]

Answer

Any two:

– **High Young's modulus** / stiff, so the extension under load is small 1

– **High breaking stress** / high tensile strength, so it can support a large load without breaking 1

– **Ductile**, so it stretches noticeably before breaking (gives a warning) 1

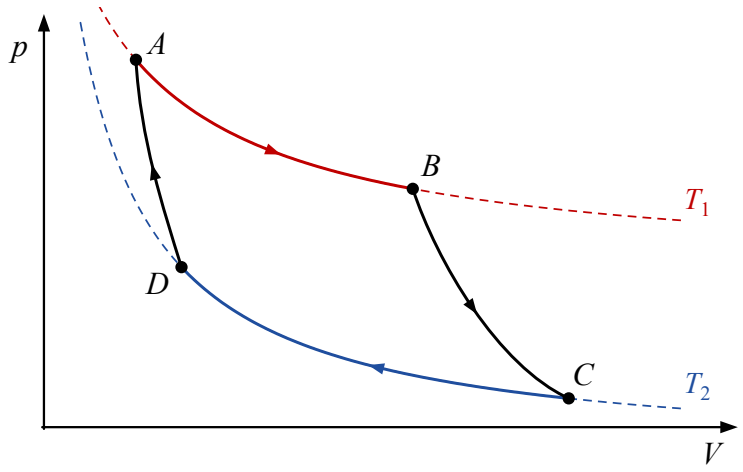
(maximum 2 marks)

2

Total for question **8**

Explanation: A lifting cable must not stretch much under load (needs a high Young's modulus) and must not snap (needs a high breaking stress). Ductility is an extra safety advantage: the cable stretches visibly before it breaks, giving a warning.

17. A thermodynamics cyclic process $ABCDA$ for 0.050 mol of a diatomic gas is shown in the p-V graph below.



The amount of heat exchanged in processes along AB and CD are 50.0 J and 25.0 J respectively. The work involved along DA is 40.0 J. The temperature difference between B and C is 50 K.

(a) Determine the work done along BC. [3]

Answer

BC is adiabatic: $Q = 0$, so $W_{BC} = -\Delta U_{BC}$ — 1

$$\Delta U_{BC} = nC_V\Delta T = (0.050)\left(\frac{5}{2}\right)(8.31)(-50) = -51.9 \text{ J}$$ — 1

$W_{BC} = 51.9$ J OR 52 J OR 51.94 J — 1

3

Explanation: The steep curves BC and DA are adiabatic: no heat is exchanged ($Q = 0$). From the first law, $W = -\Delta U$, so the gas does work by using its own internal energy, and its temperature drops by 50 K. For a diatomic gas $C_V = \frac{5}{2}R$ (3 translational + 2 rotational degrees of freedom).

(b) Determine the total work done of the cyclic process. [2]

Answer

$W = W_{AB} + W_{BC} + W_{CD} + W_{DA} = 50.0 + 51.9 + (-25.0) + (-40.0)$ (ECF from (a)) — 1

$= 36.9$ J OR 37 J OR 36.94 J — 1

(AB and CD are isothermal, so $W = Q$; DA is a compression, so $W_{DA} = -40.0$ J)

2

Explanation: The net work done in a cycle equals the area enclosed by the loop on the p–V graph. Add the work for each process with its sign: positive when the gas expands (AB, BC) and negative when it is compressed (CD, DA). For the isothermal processes $\Delta U = 0$, so $W = Q$.

(c) Determine the change in internal energy of the cyclic process. Give a reason for your answer. [2]

Answer

$\Delta U = 0$	1
The gas returns to its **initial state** / **same temperature**; internal energy depends only on the state (temperature) of the gas	1
(second mark dependent on the first)	
	2
Total for question	**7**

Explanation: Internal energy is a function of state — for an ideal gas it depends only on temperature. After a complete cycle the gas is back at state A with the same temperature, so there is no change in internal energy. This also means that, over a cycle, net heat absorbed = net work done.

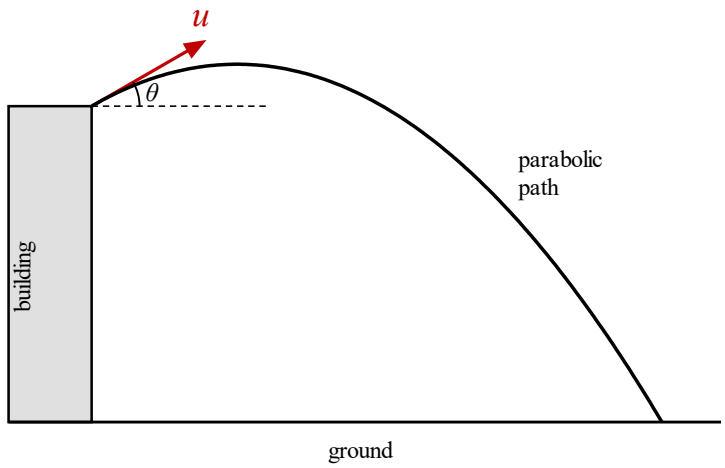
Note: ΔU = 0 for the cycle is not because the process is isothermal. Only AB and CD are isothermal, so ΔU = 0 for each of those steps. For the adiabatic steps, ΔU = −51.9 J along BC and ΔU = +51.9 J along DA. Adding all four steps: 0 + (−51.9) + 0 + (+51.9) = 0. Isothermal means ΔU = 0 for one step; a complete cycle means ΔU = 0 for the whole loop, even though some steps change U.

Section C [30 marks]

18. A man on a rooftop of a building launches a projectile with an initial velocity, u, at an angle θ above the horizontal. Assume that air resistance is negligible.

(a) Sketch the trajectory of the projectile. [2]

Answer



Parabolic path: rises from the rooftop at angle θ, then falls	1
Hits the ground **below the level of the rooftop**; u and θ labelled	1
	2

Explanation: With no air resistance, the only force is gravity. The horizontal velocity stays constant while the vertical motion has a constant downward acceleration g, so the path is a parabola. Because the projectile is launched from the rooftop, it lands below the launch level and the falling part is longer than the rising part.

(b) Derive the equation for the trajectory of the projectile. [5]

Answer

Horizontal: $x = (u\cos\theta)t$	1
Vertical: $y = (u\sin\theta)t - \frac{1}{2}gt^2$	1
$t = \dfrac{x}{u\cos\theta}$	1
$y = (u\sin\theta)\left(\dfrac{x}{u\cos\theta}\right) - \dfrac{1}{2}g\left(\dfrac{x}{u\cos\theta}\right)^2$	1
$y = x\tan\theta - \dfrac{gx^2}{2u^2\cos^2\theta}$	1
	5

Explanation: The key idea is to treat the horizontal and vertical motions separately; they are linked only by the time t. Eliminating t gives y in terms of x in the form $y = ax - bx^2$, which is the equation of a parabola.

(c) If u is 25.0 m s^{-1}, θ is 15° and the projectile hits the ground 4.0 s later, determine

(i) the height of the building, [2]

Answer

$$s_y = (u\sin\theta)t - \frac{1}{2}gt^2 = (25.0\sin 15°)(4.0) - \frac{1}{2}(9.81)(4.0)^2 = -52.6\text{ m}$$ 1

Height of building $= 52.6$ m OR 53 m OR 52.60 m 1

2

Explanation: Take the launch point as the origin and upward as positive. The vertical displacement after 4.0 s is negative, meaning the projectile is 52.6 m *below* the rooftop when it hits the ground — this is the height of the building.

(ii) the maximum height of the projectile from the rooftop, [2]

Answer

$v_y{}^2 = u_y{}^2 - 2gH$, with $v_y = 0$: $H = \frac{(25.0\sin 15°)^2}{2(9.81)}$ 1

$= 2.13$ m OR 2.1 m OR 2.134 m 1

2

Explanation: At the highest point the vertical component of velocity is momentarily zero, while the horizontal component is unchanged. The height found is measured from the rooftop, not from the ground.

(iii) the time at which the velocity is at an angle of 45° below the horizontal. [3]

Answer

$\tan 45° = \frac{|v_y|}{v_x} \Rightarrow v_y = -v_x = -25.0\cos 15° = -24.1\text{ m s}^{-1}$ 1

$v_y = u_y - gt$: $-24.1 = 25.0\sin 15° - 9.81t$ 1

$t = 3.12$ s OR 3.1 s OR 3.121 s 1

3

Explanation: The direction of the velocity is given by $\tan\alpha = \frac{v_y}{v_x}$. The horizontal component stays at 24.1 m s^{-1} throughout the flight. At 45° below the horizontal the two components are equal in size, with v_y pointing downward. Check: 3.12 s is less than 4.0 s, so this happens before the projectile lands.

(d) What will happen to the time of flight of the projectile if there is air resistance? [1]

Answer

Increases / longer (air resistance reduces the net downward acceleration during the long fall) 1

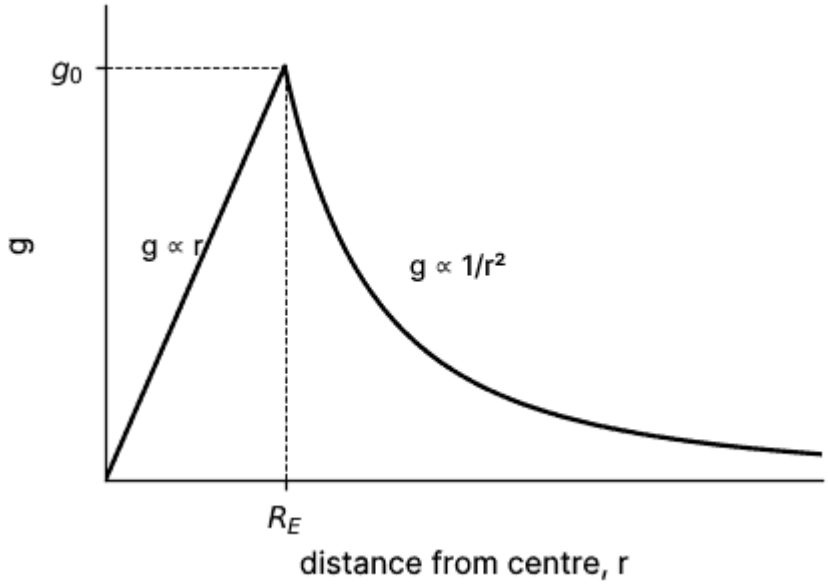
1

Total for question **15**

Explanation: Air resistance always opposes the motion. Most of this flight is a fall of about 55 m, and during the fall air resistance reduces the net downward acceleration to less than g, so the projectile takes longer to reach the ground.

19. (a) Explain, with the aid of a graph, the variations of gravitational field strength with the distance from the centre of the Earth. [4]

Answer



Graph: straight line from the origin, increasing to maximum g_0 at $r = R_E$	1
Graph: decreasing curve beyond R_E, approaching zero as r → ∞	1
Inside the Earth (uniform density): $g = \frac{GM_E r}{{R_E}^3}$, so g ∝ r	1
Outside the Earth: $g = \frac{GM_E}{r^2}$, so g ∝ $\frac{1}{r^2}$	1
	4

Explanation: Outside the Earth, Newton's law of gravitation gives $g = \frac{GM}{r^2}$, an inverse-square decrease. Inside the Earth only the mass within radius r attracts (the outer shell gives no net pull), and this mass is proportional to r^3, so $g \propto r$. The field strength is therefore greatest at the surface.

(b) A satellite that initially at rest is located 580 km above the surface of the Earth, begins to accelerate towards the Earth. Calculate the velocity of the satellite when the satellite hits the surface of the Earth. Assume that air resistance is negligible. [4]

Answer

Loss in GPE = gain in KE: $\frac{1}{2}mv^2 = GM_E m\left(\frac{1}{R_E} - \frac{1}{R_E+h}\right)$	1
$R_E + h = 6.38\times10^6 + 5.80\times10^5 = 6.96\times10^6$ m	1
$v^2 = 2(6.67\times10^{-11})(5.97\times10^{24})\left(\frac{1}{6.38\times10^6} - \frac{1}{6.96\times10^6}\right)$	1
$v = 3.23\times10^3$ m s^{-1} OR 3.2 × 10^3 OR 3.225 × 10^3 m s^{-1}	1
	4

Explanation: The value of g changes over a distance of 580 km, so the equations of uniform acceleration cannot be used. Instead use conservation of energy with gravitational potential energy $U = -\frac{GMm}{r}$. The mass of the satellite cancels out.

(c) Explain the state of weightlessness when a sky-diver jumps out of an aeroplane. [2]

Answer

The sky-diver is in **free fall**, accelerating at g; only his weight acts on him	1
There is **no normal (reaction) force**, R = 0, so his apparent weight is zero	1
	2

Explanation: Weightlessness does not mean that gravity is zero — the sky-diver's weight still acts. He feels weightless because nothing supports him: he and everything around him fall with the same acceleration, so there is no contact (normal) force. The sensation of weight comes from this contact force.

(d) A boy stands on the floor with the weight of 500 N. Then he stands on a scale in a moving lift.

(i) Determine the magnitude and direction of acceleration of the lift if the scale reading is 380 N. [4]

Answer

$m = \frac{500}{9.81} = 51.0\,\text{kg}$	1
$mg - R = ma$	1
$a = \frac{500-380}{51.0} = 2.35\,\text{m s}^{-2}$ OR 2.4 OR 2.354 m s^{-2}	1
Direction: **downward**	1
	4

Explanation: The scale reads the normal reaction R, not the true weight. If $R < mg$, the resultant force is downward, so the acceleration is downward. The lift could be moving down and speeding up, or moving up and slowing down.

(ii) State the condition when the scale in the moving lift shows the reading of 500 N. [1]

Answer

Lift moves with **constant velocity** / zero acceleration	1
	1
Total for question	**15**

Explanation: The reading equals the true weight when $R = mg$, i.e. when the acceleration is zero. A lift moving at constant velocity (up or down) gives the same reading as a stationary one — only acceleration changes the reading.

20. (a) (i) What is meant by *heat transfers through conduction*, *convection* and *radiation*? [3]

Answer

Conduction: heat transfer from a region of higher temperature to lower temperature through a medium **without bulk movement** of the medium, by **vibration of atoms** / collisions of particles (and free electrons)	1
Convection: heat transfer in a fluid by the **bulk movement of the fluid** itself, due to density differences	1
Radiation: heat transfer in the form of **electromagnetic waves (infrared)**; **no medium** is required	1
	3

Explanation: The three processes differ in how the energy is carried: by vibrating particles passing energy to their neighbours (conduction), by the fluid itself moving (convection), or by electromagnetic waves (radiation). Radiation is the only process that can occur through a vacuum.

(ii) State the difference between the mechanism of heat conductions in metals and non-metals. [1]

Answer

Metals: mainly by **free electrons** (and lattice vibrations); non-metals: only by **lattice vibrations** of atoms	1
	1

Explanation: Metals have free (delocalised) electrons that move quickly and carry energy over long distances through the lattice. This is why metals are much better conductors of heat than non-metals.

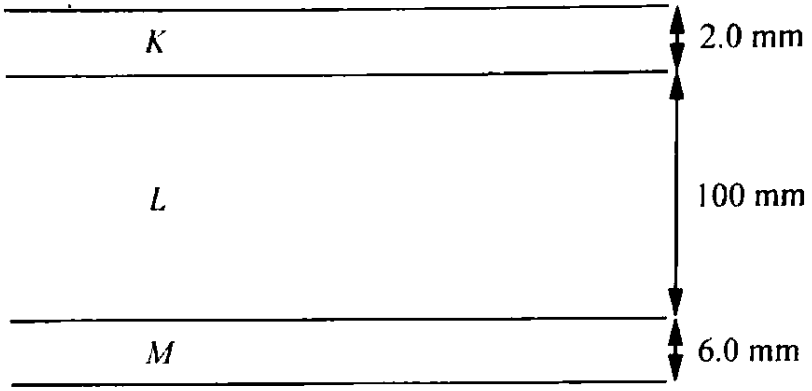
(b) State a difference between natural convection and forced convection. Give an example for each type of convection. [3]

Answer

Natural convection: fluid moves due to **density differences** caused by temperature differences; forced convection: fluid is moved by an **external device** (fan / pump)	1
Example of natural convection: sea breeze / heating water in a kettle	1
Example of forced convection: fan cooling a computer / pump circulating coolant in a car radiator	1
	3

Explanation: The key difference is what causes the fluid to move. In natural convection, warmer fluid is less dense and rises while cooler fluid sinks. In forced convection, an external device such as a fan or pump pushes the fluid.

(c) Heat transfer occurs between three layers of materials, *K*, *L* and *M*, as shown in the diagram below.



The thermal conductivities of *K*, *L* and *M* are 20 W m^{-1} K^{-1}, 0.050 W m^{-1} K^{-1} and 3.0 W m^{-1} K^{-1} respectively. If the temperatures on the top of surface *K* is 30 °C and at the bottom of surface *M* is −10 °C, calculate the rate of heat transfer per unit area. [3]

Answer

$\frac{P}{A} = \frac{\theta_1 - \theta_2}{\frac{x_K}{k_K} + \frac{x_L}{k_L} + \frac{x_M}{k_M}}$ (same rate of heat flow through each layer) 1

$$\frac{P}{A} = \frac{30 - (-10)}{\frac{0.0020}{20} + \frac{0.100}{0.050} + \frac{0.0060}{3.0}}$$ 1

$= 20.0\ \text{W m}^{-2}$ OR 20 OR 19.98 W m^{-2} 1

3

Explanation: The layers are in series: in the steady state the same rate of heat flow passes through every layer, and the temperature drops across the layers add up to 40 K. Each layer has a "thermal resistance" $\frac{x}{k}$ per unit area. Layer *L* is thick and a poor conductor, so it has by far the largest resistance and controls the rate of heat flow.

(d) A sphere light bulb of radius 3.0 mm is placed in a confined space of 25 °C. The light bulb is switched on and reaches the temperature of 2600 °C. If the emissivity of the bulb is 0.32, determine the net power radiated by the bulb. [5]

Answer

$A = 4\pi r^2 = 4\pi(3.0 \times 10^{-3})^2 = 1.13 \times 10^{-4}\ \text{m}^2$ 1

$T = 2873\ \text{K}, \quad T_0 = 298\ \text{K}$ 1

$P_{\text{net}} = e\sigma A\left(T^4 - {T_0}^4\right)$ 1

$P_{\text{net}} = 0.32(5.67 \times 10^{-8})(1.13 \times 10^{-4})(2873^4 - 298^4)$ 1

$= 1.40 \times 10^2\ \text{W}$ OR 1.4×10^2 W OR 139.8 W 1

5

Total for question **15**

Explanation: By the Stefan–Boltzmann law, the bulb emits power $e\sigma AT^4$ but also absorbs radiation from its surroundings, $e\sigma A{T_0}^4$. The net power is the difference. Temperatures must be in kelvin. Here the surroundings term is very small compared with T^4 of the hot filament.